

Design Assignment (10 marks)

Task:

You are working as a junior process engineer in a semiconductor company. The company plans to fabricate a low-power CMOS inverter (nMOS + pMOS on the same wafer) for a digital IC operating at 1.8 V. Your task is to design a feasible monolithic fabrication process flow, starting from a silicon wafer and ending at a packaged chip under Practical Constraints

You must briefly:

1. Give a step-by-step process flow (core design)

- ❖ Wafer preparation and cleaning (incl. one RCA sequence).
- ❖ Thermal oxidation (field and gate oxides, with target thicknesses).
- ❖ Photolithography steps for all mask levels (coat, bake, expose, develop, align).
- ❖ Doping (diffusion/ion implantation) for wells and source–drain.
- ❖ Etching (at least one wet and one dry; note selectivity/anisotropy).
- ❖ Film deposition (e.g., polysilicon, metal, passivation: evaporation/sputter/CVD).
- ❖ Isolation method (p–n, LOCOS/oxide, or STI – simplified).
- ❖ Contact, metallization, passivation, basic testing, bonding & packaging.

Rubric (10 marks)

Criteria	Weight (Marks)	Excellent (Full marks)	Satisfactory (Half marks)	Poor (0 marks)
Problem Understanding & Specification	2	Requirements, device function, and constraints are clear, consistent, and realistic.	Basic function stated, some constraints mentioned, but incomplete/unclear.	Requirements are largely missing or incorrect.
Process Flow Design & Technical Correctness	3	Complete, logical CMOS process including all key steps; technically sound and consistent with course content.	Mostly correct with minor omissions/errors (e.g., missing one major step or minor technical slip).	Flow is highly incomplete or wrong; major steps are missing or misused.
Design Justification	3	Clear justification of key choices (oxide, junction depth, methods)	Some justification; discussion general or shallow.	Little/no justification;

Presentation & Diagrams	2	Well-structured, concise report; process table and sketches neat and readable; technical terms used correctly.	Understandable but with noticeable issues in organization, clarity, or figures.	Disorganized, hard to follow; missing or unclear sketches/tables.
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